Brayton Cycle Formulas

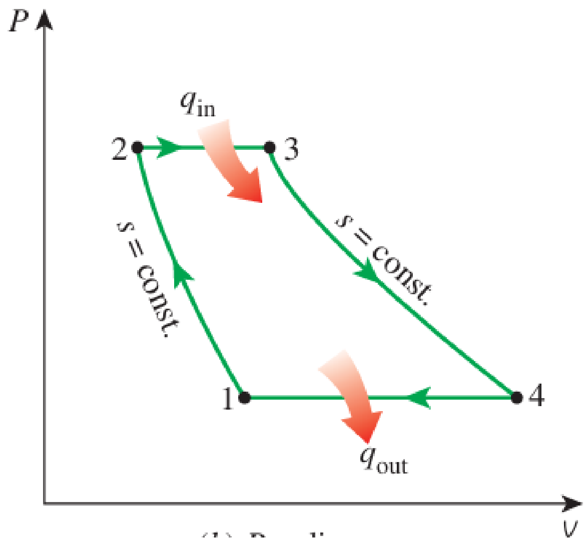

Figure 1: Brayton Cycle P-v Diagram

1. Thermal Efficiency

$$\eta_{th} = \frac{W_{net}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

Heat Transfer:

$$Q_{in} = C_p(T_3 - T_2), \qquad Q_{out} = C_p(T_4 - T_1),$$

so

$$\eta_{th} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

Using isentropic relations:

$$\frac{T_2}{T_1} = \frac{T_3}{T_4} \gg \frac{T_4}{T_3} = \frac{T_1}{T_2}$$

Substitute

$$\eta_{th} = 1 - \frac{T_1}{T_2}$$

$$\frac{T_2}{T_1} = r_p^{\frac{k-1}{k}}$$

$$\boldsymbol{\eta_{th} = 1 - \frac{1}{r_p^{\frac{k-1}{k}}}}$$

2. Back Work Ratio

$$BWR = \frac{W_c}{W_t}$$

Work Expression:

$$W_c = C_p(T_2 - T_1), \qquad W_t = C_p(T_3 - T_4),$$

so

$$BWR = \frac{T_2 - T_1}{T_3 - T_4}$$

Using isentropic relations:

$$T_2 = T_1 * r_p^{\frac{k-1}{k}}$$

$$T_4 = \frac{T_3}{r_p^{\frac{k-1}{k}}}$$

$$\boldsymbol{BWR = \frac{r_p^{\frac{k-1}{k}} - 1}{\frac{T_3}{T_1} - r_p^{\frac{k-1}{k}}}}$$

EES Code

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"Spring 2026 - Applied Thermodynamics Project"

"Atmospheric Pressure and Temperature at Inlet"

T_1 = 300 [K]

P_1= 100 [kPa]

r_p = 10

P_2 = r_p*P_1

P_1=P_4

P_2=P_3

"Turbine Inlet Temperature"

T_3 = 1000 [K]

"Specific Heat Ratio"

"k = 1.3"

"Brayton Cycle Efficiency"

eta_Th = 1-(1/r_p^((k-1)/k))

"Back Work Ratio for Brayton Cycle"

BWR =(r_p^((k-1)/k)-1) / ((T_3/T_1)- (r_p^((k-1)/k)))
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EES Plots

Efficiency and Back Work Ratio Both increases with increase in specific heat capacity for Brayton cycle as shown in Figure 2 and Figure 3.

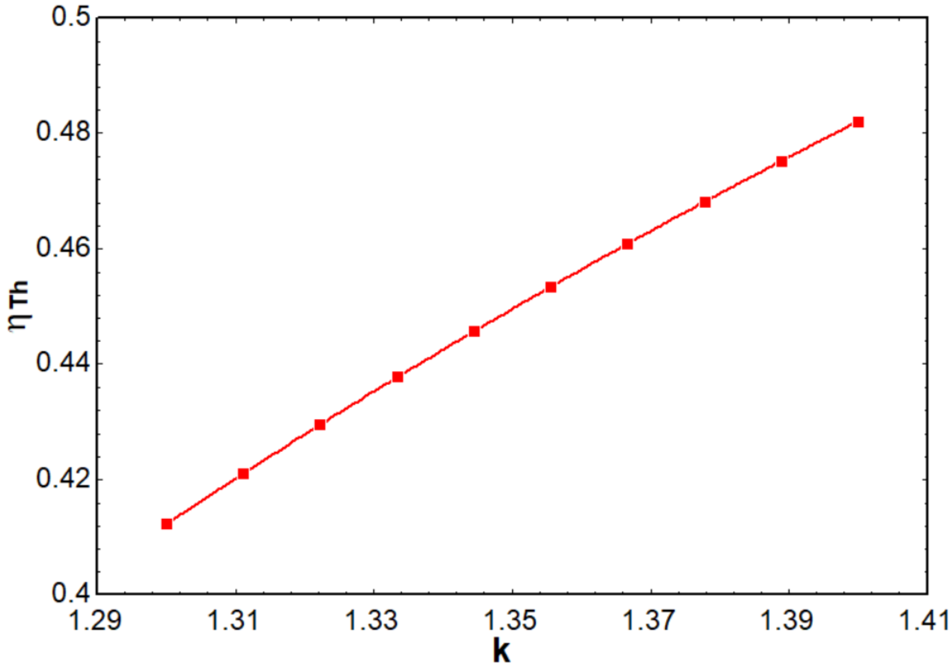

Figure 2: Thermal Efficiency Vs Specific Heat Ratio

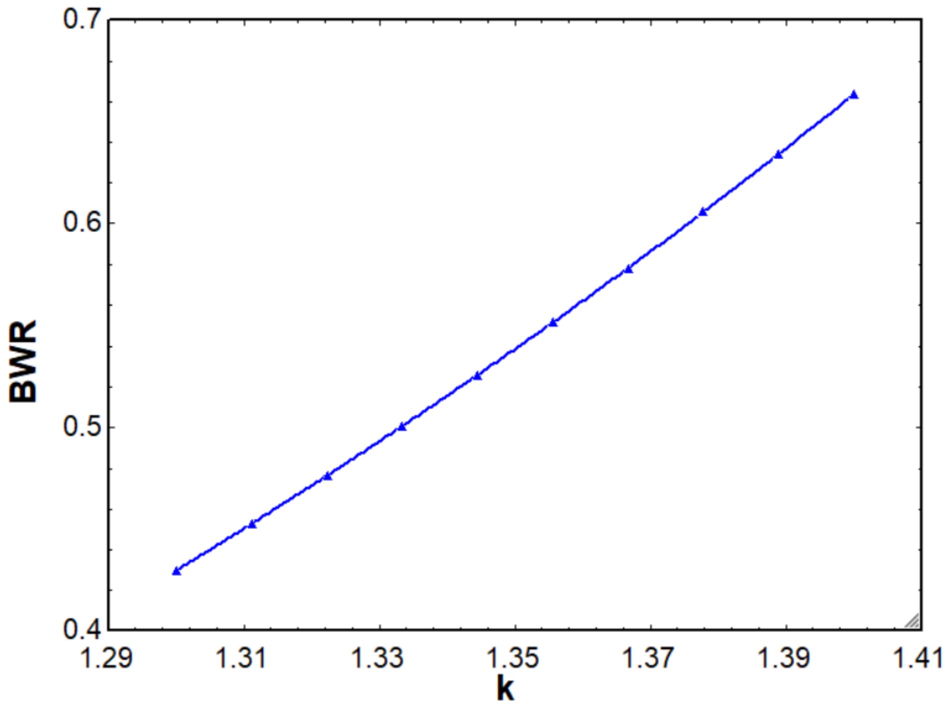

Figure 3: Back Work Ratio Vs Specific Heat Ratio